

GREAT FALLS, MB, Canada

WMO: 715700

Lat: 50.459N

Lon: 96.011W

Elev: 830

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.1	-21.2	-33.5	0.9	-25.8	-28.3	1.2	-20.9	15.9	10.6	14.3	10.1	3.7	170	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	84.4	66.9	81.4	65.3	78.8	63.7	70.3	79.6	68.3	77.7	66.4	75.2	7.1	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.1	102.6	76.1	64.9	95.0	73.7	62.9	88.5	71.5	34.9	79.8	33.1	77.8	31.6	75.0	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.1	13.6	12.1	DB	-32.6	90.0	4.8	3.1	-36.0	92.2	-38.8	94.0	-41.5	95.8	-44.9	98.0	(4)
			WB	-32.8	74.3	4.7	2.7	-36.2	76.2	-39.0	77.8	-41.6	79.3	-45.1	81.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	4.4	5.5	22.2	38.6	51.7	62.4	67.6	65.6	57.1	43.1	26.3	9.7	(6)
(7)		DBStd	25.02	15.09	13.36	13.82	10.89	8.90	6.20	5.67	5.62	7.88	8.67	11.80	13.31	(7)
(8)		HDD50	6207	1413	1245	865	359	87	2	0	0	22	253	711	1250	(8)
(9)		HDD65	10112	1878	1665	1328	791	420	117	38	61	259	680	1160	1715	(9)
(10)		CDD50	1838	0	0	2	18	139	375	545	485	234	39	1	0	(10)
(11)		CDD65	270	0	0	0	0	7	40	119	81	22	1	0	0	(11)
(12)		CDH74	2036	0	0	0	1	102	323	887	582	137	4	0	0	(12)
(13)		CDH80	410	0	0	0	0	15	46	206	122	21	0	0	0	(13)
(14)	Wind	WSAvg	5.9	5.5	5.3	5.9	6.5	6.5	5.7	5.4	5.4	6.1	6.5	6.4	5.5	(14)
(15)	Precipitation	PrecAvg	18.5	0.7	0.5	0.8	1.1	2.1	3.3	2.7	2.6	2.2	1.3	0.9	0.7	(15)
(16)		PrecMax	23.5	2.7	1.9	1.7	2.1	6.4	9.6	6.1	5.3	4.3	4.4	3.4	1.9	(16)
(17)		PrecMin	11.3	0.0	0.0	0.0	0.0	0.3	0.3	1.1	0.4	0.2	0.0	0.0	0.1	(17)
(18)		PrecStd	3.5	0.6	0.4	0.5	0.6	1.4	2.1	1.3	1.4	1.2	0.9	0.8	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.1	39.8	59.7	70.4	81.6	84.9	89.6	87.6	83.0	72.4	57.4	37.9	(19)
(20)		MCWB	33.9	34.7	47.7	51.2	59.1	65.6	71.6	68.6	65.9	57.7	48.6	33.5	(20)	
(21)		2%	DB	33.8	34.7	50.1	64.3	76.8	80.7	85.2	83.3	77.9	66.3	50.4	33.3	(21)
(22)		MCWB	31.7	31.7	41.8	48.5	56.7	63.5	68.5	66.2	63.5	54.5	43.3	30.9	(22)	
(23)		5%	DB	30.3	30.5	44.4	59.6	71.7	77.7	81.7	80.1	74.0	60.7	45.6	31.1	(23)
(24)		MCWB	28.4	27.4	37.6	45.7	54.7	61.4	66.2	65.7	61.3	51.3	40.0	29.2	(24)	
(25)		10%	DB	26.0	25.2	39.7	55.4	66.8	74.9	78.9	77.0	69.7	55.9	41.2	28.2	(25)
(26)		MCWB	24.2	23.0	34.9	43.7	53.4	61.0	65.0	63.9	60.1	48.9	37.4	26.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.5	35.1	50.8	52.2	64.1	70.7	74.6	71.9	69.1	59.0	48.9	33.7	(27)
(28)		MDCB	36.4	38.5	54.5	65.9	71.4	78.8	86.2	81.8	77.1	70.7	56.0	37.1	(28)	
(29)		2%	WB	31.7	32.1	42.2	49.6	60.7	66.9	71.4	69.5	66.0	55.7	43.9	31.5	(29)
(30)		MDCB	34.0	34.6	49.6	61.3	68.5	75.5	80.7	79.3	73.6	64.2	49.2	33.3	(30)	
(31)		5%	WB	28.3	27.5	38.2	47.1	57.6	64.5	69.0	67.5	63.6	52.4	40.3	29.2	(31)
(32)		MDCB	30.2	30.2	43.6	57.3	67.8	73.2	78.5	77.0	71.3	59.8	44.7	31.0	(32)	
(33)		10%	WB	24.3	23.0	35.2	44.7	55.2	62.5	66.9	65.7	61.1	48.9	37.4	26.4	(33)
(34)		MDCB	25.9	25.4	39.1	54.8	64.6	71.1	76.0	74.2	68.0	55.4	41.0	28.3	(34)	
(35)	Mean Daily Temperature Range	MDBR	17.6	20.2	20.3	19.8	19.2	17.8	18.2	17.8	16.7	12.9	11.4	13.6	(35)	
(36)		5% DB	MCDBR	16.1	20.2	20.0	25.4	26.9	22.8	21.4	22.3	21.7	20.0	16.3	12.7	(36)
(37)		MCWBR	14.5	16.7	13.2	13.5	12.1	10.2	9.7	10.2	10.2	11.2	10.5	11.0	(37)	
(38)		5% WB	MCDBR	15.3	18.7	18.7	22.3	20.4	18.0	18.6	18.7	18.1	18.2	14.7	12.0	(38)
(39)		MCWBR	13.9	16.0	12.7	12.7	11.0	9.8	10.2	10.0	10.0	10.8	10.0	10.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.257	0.280	0.291	0.340	0.354	0.377	0.385	0.370	0.341	0.301	0.274	0.248	(40)	
(41)		taud	2.270	2.216	2.341	2.393	2.428	2.388	2.385	2.442	2.508	2.582	2.477	2.331	(41)	
(42)		Ebn at Noon	249	271	291	287	286	279	275	274	272	267	246	237	(42)	
(43)		Edh at Noon	21	30	32	34	35	36	36	32	27	21	18	17	(43)	
(44)	All-Sky Solar Radiation	RadAvg	380	746	1149	1591	1709	1791	1879	1558	1113	645	347	276	(44)	
(45)		RadStd	39	56	81	116	138	146	99	117	101	82	45	29	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.22	N/A	N/A	N/A	N/A

Nomenclature: See separate page